

Optimal Solar Panel Tilt & Azimuth determination

FOUNDATIONS OF
ARTIFICIAL INTELLIGENCE

Presented by

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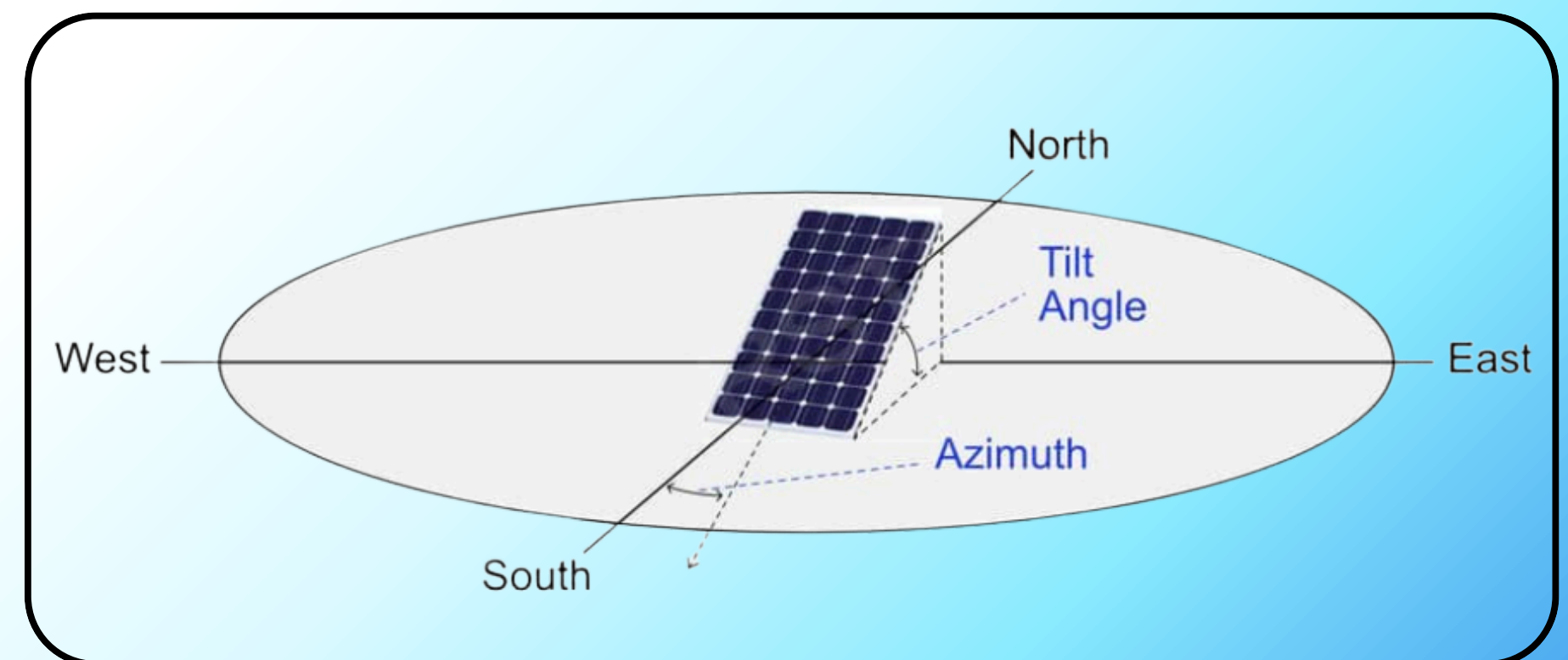




Problem Statement

The energy generated by a solar photovoltaic (PV) panel depends on its tilt and azimuth angles, which vary according to the geographical location. Selecting suitable angles manually can be time-consuming and may not provide the maximum possible annual energy yield.

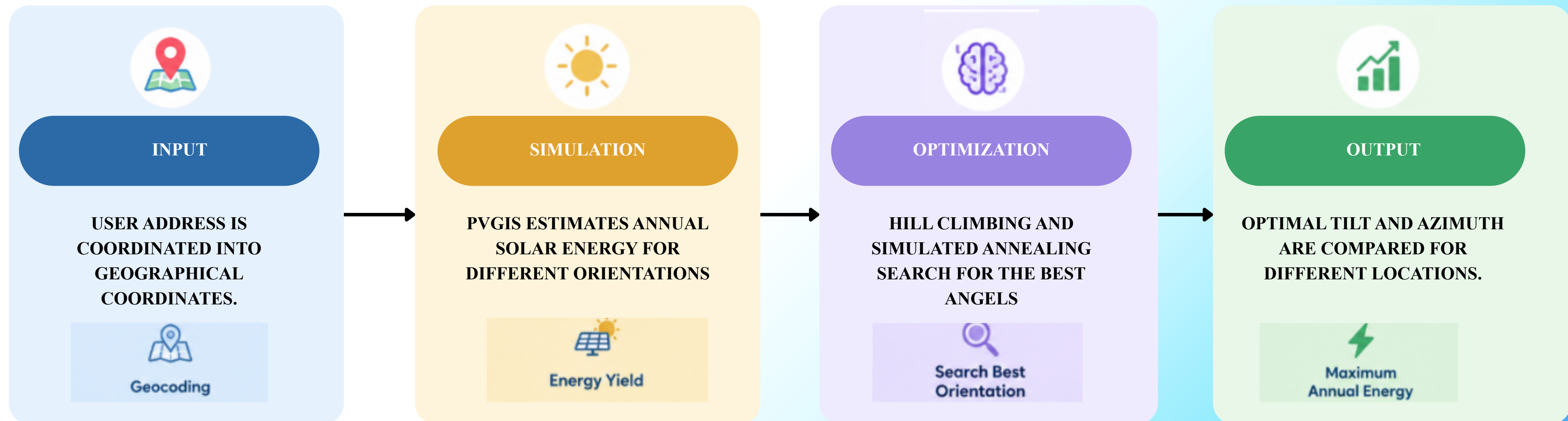
Therefore, there is a need for a system that can automatically determine the optimal tilt and azimuth angles for a given location. The proposed system uses geocoding to obtain the location coordinates, PVGIS to estimate annual solar energy yield, and Hill Climbing and Simulated Annealing algorithms to search for an orientation that maximizes energy generation.





Abstract

- The proposed project aims to determine the optimal tilt angle and azimuth angle of a solar photovoltaic (PV) panel for a given geographical location in order to maximize its annual energy generation. Instead of manually testing a large number of possible panel orientations, the problem is formulated as a local-search optimization problem.
- The geographical location is obtained from the user's address through geocoding, while the PVGIS API provides the predicted annual energy yield for different panel orientations.
- Hill Climbing and Simulated Annealing algorithms are used to search for an optimal combination of tilt and azimuth.





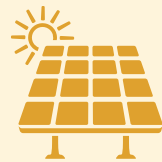
Literature Survey

- Solar PV output depends significantly on geographical location, tilt and azimuth angles.
- Existing research uses solar simulation models and energy databases to estimate pV energy generation for different geographical locations.
- Various optimization techniques such as Genetic Algorithms, particle Swarm Optimization and other search based methods have been explored to improve solar panel optimization.
- Our work applies Hill Climbing and Simulated Annealing with real PVGIS energy yield data to optimize both tilt and azimuth for a user selected location.



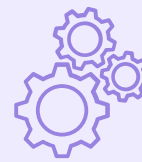
EXISTING RESEARCH

STUDIES ON SOLAR PV
ORIENTATION AND ENERGY
GENERATION



SOLAR SIMULATION

USE OF SIMULATION TOOLS
AND SOLAR ENERGY
DATABASES



OPTIMIZATION TECHNIQUES

E.G., GENETIC ALGORITHMS,
PARTICLE SWARM
OPTIMIZATION, SEARCH
BASED METHODS



IDENTIFIED GAPS

MOST APPROACHES
EVALUATE PRE-DEFINED
ORIENTATIONS OR USE
SINGLE TECHNIQUES



PROPOSED AI BASED APPROACH

HILL CLIMBING AND
SIMULATED ANNEALING
WITH REAL PVGIS ENERGY
YIELD DATA

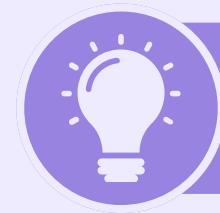


Existing Solution vs Proposed Solution



Existing Solution

- Panel tilt and azimuth are often selected manually or based on general guidelines.
- Predefined panel orientations are tested to identify a suitable angle.
- Uses fixed assumptions or simulation-based values.
- Limited adaptation to specific geographical locations.
- Testing many possible orientations can be time-consuming.
- Does not typically compare multiple optimization techniques.
- The final orientation may not provide the maximum possible annual energy yield.



Proposed Solution

- User enters the location/address.
- Geocoding automatically converts the address into latitude and longitude.
- PVGIS provides real solar-energy yield data for the selected location.
- Hill Climbing searches for improved tilt and azimuth combinations.
- Simulated Annealing explores the search space and helps overcome local optima.
- The system automatically determines the optimal tilt and azimuth.
- The maximum predicted annual energy yield is displayed.

From manual estimation to intelligent optimization



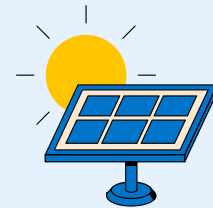
Research Gap

1 What's already known

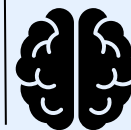
- Solar Panel optimization has been studied before.
- Many conventional orientations approaches evaluate different orientations directly.
- These methods often require manual testing of multiple tilt and azimuth angles.

2 The Opportunity

The real gap lies in combining existing solar energy optimization with local search methods in a practical, location-dependant system.



Existing Solar
Energy
Optimization



AI Local
Search
Methods



Practical,
Location Dependent
System

3 The Problem Formulation



State



Neighbouring States



Objective Function



Search Strategy

4 Investigating Location Dependency



The project also compares how the optimal orientation changes across different geographical locations by studying cities such as Chennai, Bengaluru and Leh.

5 Identified Research Gap



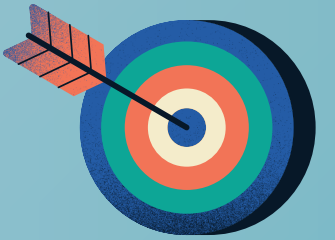
The proposed work investigates the practical application and comparative performance of Hill Climbing and Simulated Annealing for optimizing both Solar Panel tilt and Azimuth using real world PVGIS energy-yield data while also studying how the resulting optimal orientation varies across different geographical locations.



Research Objective

Main objective

- Obtain the latitude and the longitude of a users location from their address using geocoding.



1



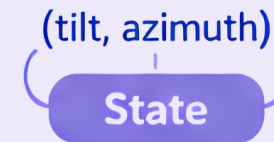
- Obtain the latitude and the longitude of a users location from their address using geocoding.

2



- Use the PVGIS API to obtain annual solar-energy yield for different panel orientations.

3



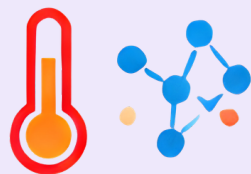
- Formulate solar-panel orientation as a local-search optimization problem where (tilt, azimuth) represents a state.

4



- Implement Hill Climbing to search for an improved panel orientation.

5



- Implement Simulated Annealing to explore the search space and reduce the possibility of getting trapped in a local optimum.

6



- Compare the performance and results of Hill Climbing and Simulated Annealing.

7



- Compare optimized panel orientations across Chennai, Bengaluru, and Leh to investigate the influence of geographic location and latitude.

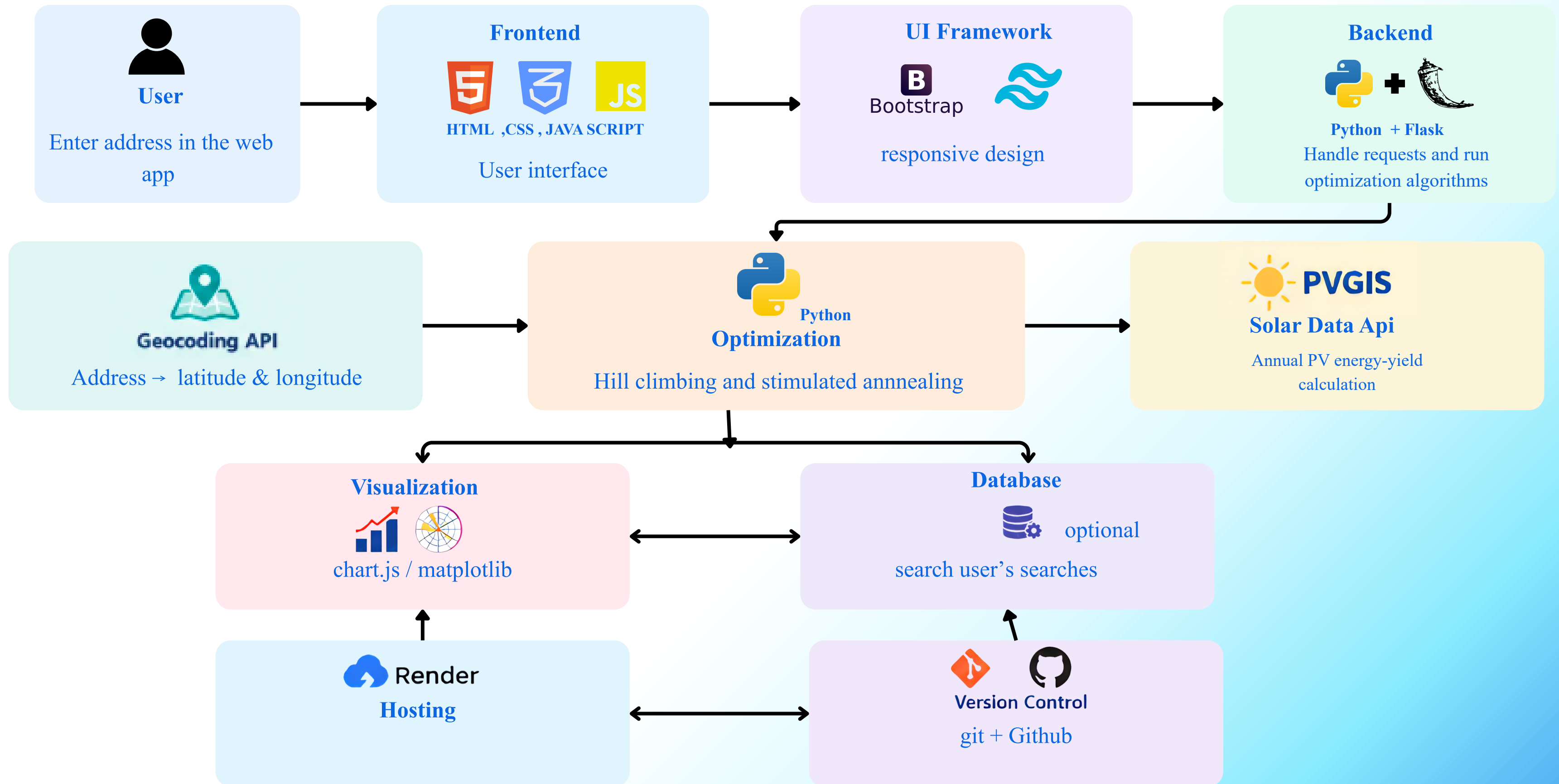
8



- Identify the orientation that provides the highest predicted annual energy yield for the selected location.



Architecture





Algorithms

HILL CLIMBING AND SIMULATED ANNEALING



Problem Statement

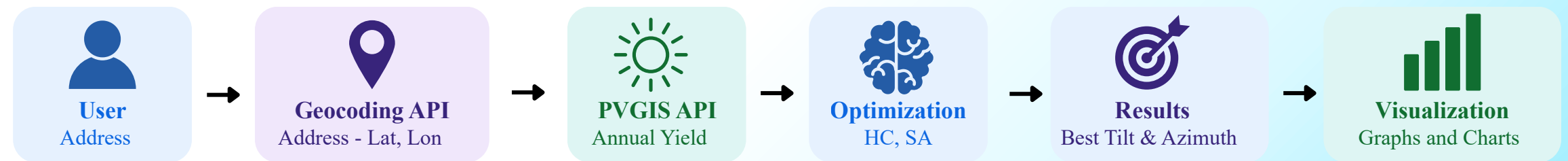
Given a real geographical location, find the tilt angle and azimuth for a solar panel that maximises total annual energy yield using real solar irradiance data from the EU's PVGIS API, solved through local search algorithms, (Hill Climbing, then Simulated Annealing) instead of brute force testing.



Why this is Unique?

- Treats the problem as a search function (state, action, object function)
- Compares two local search algorithms
- Studies location based variation
- Real PVGIS data
- Explores whether Simulated Annealing provides an advantage over Hill Climbing

OVERALL WORKFLOW



Hill Climbing

```
def hill_climb(lat, lon, start_tilt=20,
              start_azimuth=0, step=2, max_iters=100):
    tilt, azimuth = start_tilt, start_azimuth
    current_yield = get_yield(lat, lon, tilt, azimuth)
    for i in range(max_iters):
        neighbors = [(tilt + step, azimuth), (tilt - step, azimuth),
                    (tilt, azimuth + step), (tilt, azimuth - step)]
        neighbors = [(max(0, min(90, t)), max(-90, min(90, a)))
                    for t, a in neighbors]
        best_neighbor, best_yield = None, current_yield
        for t, a in neighbors:
            y = get_yield(lat, lon, t, a)
            if y > best_yield:
                best_yield, best_neighbor = y, (t, a)
        if best_neighbor is None:
            break # converged
        tilt, azimuth = best_neighbor
        current_yield = best_yield
    return tilt, azimuth, current_yield
```

Simulated Annealing

```
def simulated_annealing(lat, lon, start_tilt=20,
                       start_azimuth=0, initial_temp=100,
                       cooling_rate=0.95, max_iters=100):
    tilt, azimuth = start_tilt, start_azimuth
    current_yield = get_yield(lat, lon, tilt, azimuth)
    best_tilt, best_azimuth, best_yield = tilt, azimuth, current_yield
    temp = initial_temp
    for i in range(max_iters):
        new_tilt = max(0, min(90, tilt + random.randint(-3, 3)))
        new_azimuth = max(-90, min(90, azimuth + random.randint(-3, 3)))
        new_yield = get_yield(lat, lon, new_tilt, new_azimuth)
        delta = new_yield - current_yield
        if delta > 0 or random.random() < math.exp(delta / temp):
            tilt, azimuth, current_yield = new_tilt, new_azimuth, new_yield
            if current_yield > best_yield:
                best_tilt, best_azimuth, best_yield = tilt, azimuth, current_yield
        temp *= cooling_rate
    return best_tilt, best_azimuth, best_yield
```

Multi-City Comparison

```
cities = {
    "Chennai": (13.0827, 80.2707),
    "Bengaluru": (12.9716, 77.5946),
    "Leh": (34.1526, 77.5771)
}

results = {}

for city, (lat, lon) in cities.items():
    tilt, azimuth, yield_val, _ = simulated_annealing(lat, lon)
    results[city] = (tilt, azimuth, yield_val)

print(f'{city}: tilt={tilt}°, azimuth={azimuth}°, '
      f'yield={yield_val:.1f} kWh')
```



Implementation



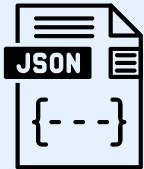
Programming Language Python

Used for communicating with the PVGIS API, implementing the optimization algorithms and processing the JSON from data.



Requests Library

Used to send HTTP requests to the PVGIS API and handle the response.



getJSON Processing

The returned JSON from data is parsed to extract the annual energy yield.



API

EU Joint Research Centre PVGIS API

Provides solar energy yield data for different panel orientations.



Geocoding

Uses a geocoding API (e.g., [geocode maps.co](https://geocode.maps.co/)) to convert the user's address into latitude and longitude.



Key Configuration



Peak Power

1 kW



System Loss

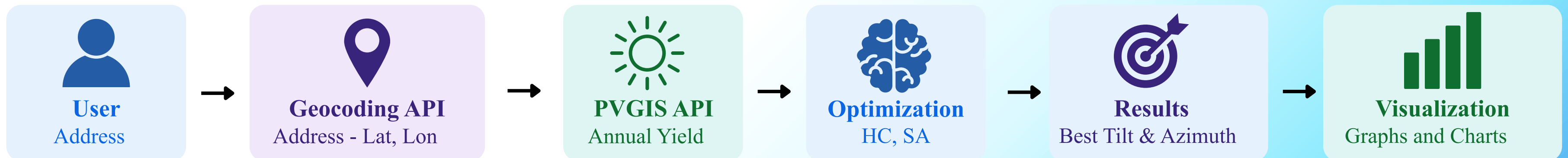
14%

The optimization algorithm determines the tilt and azimuth. The user only provides the location, not these values.

Platform

Python application (local or cloud-based).
Can be deployed using **Render** for web access.

SYSTEM WORKFLOW





Output



SolarOpti

Home

About

How It Works

Results



Better Orientation
Greater Energy

Optimize Your Solar Panel Orientation

Enter your location and let our AI-powered algorithms find the best tilt and azimuth for maximum annual energy yield using real PVGIS data.

Enter Your Address

Chennai, Tamil Nadu, India

Find Location

Location found!

Chennai, Tamil Nadu, India
Lat: 13.0827 Lon: 80.2707

Optimization Settings

Peak Power 1 kW (fixed)

System Loss 14% (fixed)

Output Format JSON (fixed)

You only need to provide the location.
The system automatically handles the rest.

Get Optimal Orientation

Optimal Orientation Found!



Optimal Tilt Angle

20°



Optimal Azimuth Angle

0°

(South-facing)



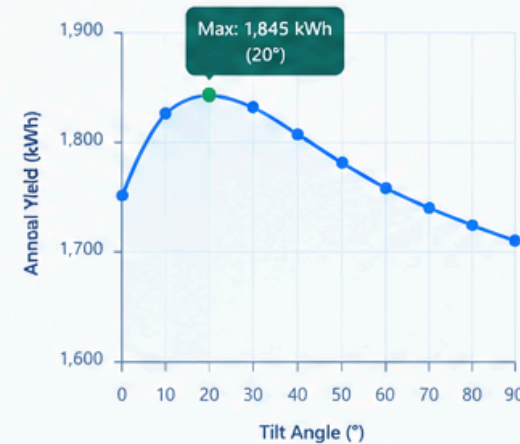
Predicted Annual Energy Yield

1,845 kWh

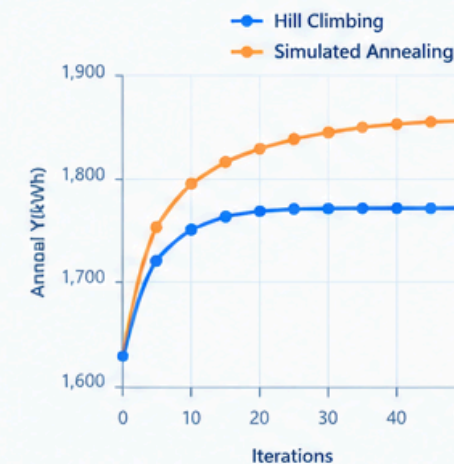
(for 1 kW panel)



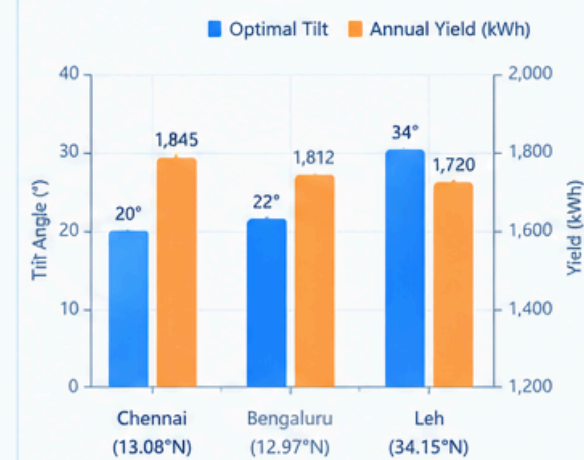
Yield vs Tilt Angle (Azimuth = 0°)



Algorithm Convergence



Comparison Across Cities



Smart Orientation • Higher Energy • A Greener Tomorrow

Powered by PVGIS | Built with Python & Flask

